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Essay / Assignment Title: Relational Database Design, Implementation and Analytics Case Study of a Telecommunications Provider

Programme title: Msc Data Analytics

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CONTENTS

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INTRODUCTION

Prior to emergence of data-driven decision making, organizations depended on database technologies to store, process as well as analyze huge amounts of data on operations. The report is a proposal to design and implement an end-to-end relational database system of a telecommunications service provider, a field where data is an important part of day-to-day operations and strategy analysis.

The project is then divided into five chapters each of which deals with a major stage in database development.

Chapter 1 is devoted to defining the requirements of data in operations of the telecommunications industry and converting them into a complete relational schema in normal form. This involves the definition of various entities, many-to-many relationships with the help of junction tables and the application of integrity constraints with the help of SQL.

Chapter 2 is about data population data and development of analytical SQL views. Realistic sample data is then entered to model operational situations and database views are created to summarize performance as well as trend analysis.

The scope of the analysis is enlarged in Chapter 3 as complex SQL queries are applied with the joins, aggregations, subqueries, CASE expressions, and window functions. It also presents a user-defined function and a stored procedure to illustrate procedural SQL.

The fourth chapter studies how the CAP Theorem can be applied to the distributed telecommunications system, and in it the trade-offs between consistency, availability, and partition tolerance are addressed.

Lastly, Chapter 5 covers query performance optimization whereby a query with poor performance is analyzed and optimization techniques of indexing and query reform are applied.

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https://drive.google.com/drive/folders/1EiyLSZ4ra0_XDSpo61ni9h9T3uNigNYH?usp=sharing

CHAPTER ONE: Relational Database Design and Normalization for the Telecommunications Provider

1.1 Domain Selection and Overview

The selected area to be covered under this project is one of the Telecommunications Service Providers, an industry that manages massive amounts of structured information concerning customers, subscriptions, billing, usage and customer services. The design principles used in the given project adopt the existing methodologies of database discussed by Connolly and Begg (2015), Elmasri and Navathe (2016) and Harrington (2016) stating the significance of normalization, integrity constraints, and relational modelling. Moreover, the telecom environment operational requirements are related to the requirements of the telecom industry that are described by the GSM Association (GSMA, 2022) and emphasize the importance of precise subscriber data management, detailed usage tracking, and scalable service arrangements.

Executable Script: Mysql_Meadhavi.sql

Operational Data needs of the Telecom Provider:

Telecommunications providers process high amounts of structured, semi structured and transactional information. In this project, the data requirements are those related to the core business processes required to provide mobile network services.

Customer management will entail the storage of sensitive user data on the customers to facilitate account creation, subscription activation, billing and customer care. It contains personal information, billing and service addresses, customer registration and history of status.

Service and subscription administration encompasses mobile plans, subscriptions and optional add-on services e.g. additional data packages. The important information is in the form of plan price, voice and data allowance, service option and subscription information like MSISDN, SIM number, dates of activation and changes of status. Add-on goods and the recurring fees are also handled.

Telecommunication networks capture the records of call details (call time, duration, destination and charges) and the mobile data sessions such as session duration, data usage and billed. The records are used to invoice, detect fraud and analyse customer behaviour.

Billing and payment processing is in charge of subscription or prepaid billing cycles as well as customer payment records.

Operational data includes:

Billing data encompasses invoice periods, issue and due date, amounts, mode of payment and payment status (partially or delayed).

Time-based support tickets, type of issues, priorities, status of resolution and assignment of an agent are captured in customer support operations including cases of multiple agents.

Product and feature management deals with reusable network features like roaming, 5G access, and streaming benefits and there are many-to-many relationships between plans and features.

Reasons why this domain is suitable for a Relational Database:

The telecommunications sector is among the areas that best accommodate the use of a relational database because the data is in large size and structured, the accuracy is paramount and also relational query is essential to analyze the billing and usage. It has also several many to many associations, i.e., between plans and features, customers and addresses, and support tickets and support agents.

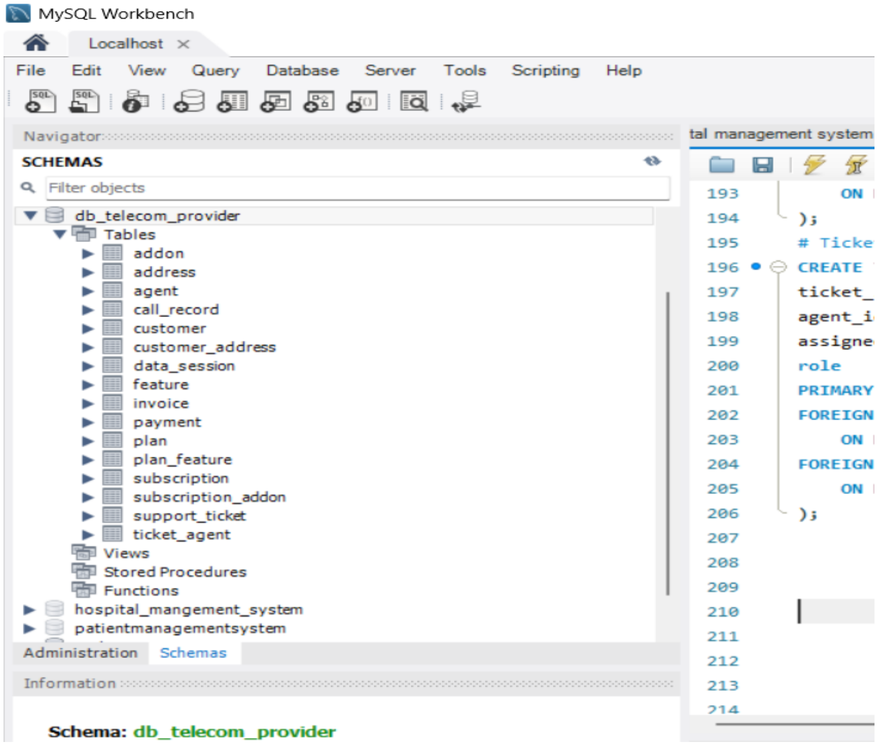
1.2 Schema Implementation:

In this section, the SQL implementation of the telecom database schema has been presented in terms of the tables with the corresponding key and constraints. Many-to-many relationships are addressed using junction tables in order to have a normalised relational design.

ORDER OF TABLE CREATION:

1. INDEPENDENT TABLES (no FKs)
    - Customer
    - Address
    - Plan
    - Feature
    - Add On
    - Agent
2. Junction tables for M:N relationships:
    - Customer_Address
    - Plan_Feature
3. Dependent tables with foreign keys:
    - Subscription
    - Invoice
    - Payment
    - Call_Record
    - Data_Session
    - Support_Ticket
4. Junction tables depending on dependent tables:

- Subscription_AddOn
- Ticket_Agent



1.3 Identifying the entities and relationships designing:

Entity Identification:

Following the domain analysis of the telecommunications provider, sixteen entities were selected as important to support the operations, billing, usage and support processes. These are the customer, service, plan, usage, payment and support facilities. The entities bear another set of attributes and supplement the general relational structure of the database.

The main entities are described below:

1. Customer: Represents individuals or businesses who are subscribers to telecom services. Stores personal/contact information and customer status.

2. Address: Physical address of the stores where to bill and deliver services, shipments or correspondence.

3. Plan: Represents mobile tariff/subscriptions plans that have price, allowances (minutes, data), plan-type details.

4. Feature: Represents optional plan features, e.g., roaming, 5G or international minutes.

5. AddOn: Represents other services that can be purchased by the customers as an addition to a subscription (e.g. extra data bundle).

6. Subscription: Ratified by a customer to a mobile line or service such as MSISDN (phone number), SIM data and plan choice.

7. Invoice: Represents records of billing produced each billing cycle on a per-subscription basis.

8. Payment: Carries payment information on invoices such as method and amount as well as status of the transaction.

9. Call_Record: Store call details records (CDRs) containing timestamps, duration, call type and charges.

10. Data_Session Stores mobile data consumption history (MB used, session time and charges).

11.Support_Ticket: Tracks Customer support addresses customer problems in billing, network, technical or any general questions.

12. Agent Representatives assist agents whose responsibility is to solve tickets.

13. Customer Number, Address (junction table): Resolves the M:N quasi association between Addresses and Customers.

14. Plan_Feature (junction table) Relationships Plans and Features: resolves the M:N relationship.

15. Subscription_AddOn (join table) M:N Solution between Subscriptions and AddOns.

16. Ticket_Agent (junction table) Relationship: Support Tickets and Agents: M:N Relationship.

Relationship Design:

The entity-entity relationships are indicative of telecom entities activities in the real world. These relationships are categorized into three namely one-to-many (1:N), many-to-one (N:1) and many-to-many (M:N).

A. One to Many (1:N) Relationships:

1. Customer → Subscription: A single customer may be subscribing to more than one subscription (mobile line). Every subscription is associated with a single customer.

2. Subscription → Invoice: Every subscription will produce several invoices within billing cycles.

3. Invoice → Payment: A payment can be on a single invoice or it can be in partial payments.

4. Subscription → Call_Record: A subscription may give rise to a large number of call detail records.

5. Subscription → Data_Session: A subscription provides numerous sessions of data usage.

6. Subscription → Support_Ticket: There can be several support tickets raised on a subscription.

B. Many to Many(M:N) Relationships:

Relationship 1: Customer to Address:

Several addresses to a single customer are possible. (e.g., billing, shipping, service) and one and the same address can belong to several customers. (e.g. shared household, office building).

Resolved by Junction table Customer_Address

Relationship 2: Plan to feature:

Each plan may offer a number of features and a feature may be in a number of plans.

Resolved by Junction table Plan_Feature

Relationship 3: Subscription to AddOn:

A subscription can be extended to have several services (e.g. additional 5GB data + global minutes) and a subscription can be taken to an addition by a multitude of clients.

Relationship 4: Support_Ticket to Agent:

Multiple agents (primary, secondary, escalation) can deal with support tickets. There are a lot of tickets that can be worked upon by agents.

Resolved by Junction table Ticket_Agent

It has 16 interrelated objects that assist customer-level, subscription level, billing level, usage level and support level operations in telecommunications database. One-to-many and many-to-many relations describe customers, plans, features, add-ons, addresses and support agents relations and make them flexible.

1.4 ENTITY-RELATIONSHIP DIAGRAM (ERD)

Entity-Relationship Diagram (ERD) is the entire logical form of the database of the telecommunications provider. It displays all objects, primary keys, foreign key constraints and the cardinality between objects with crow’s-foot notation.

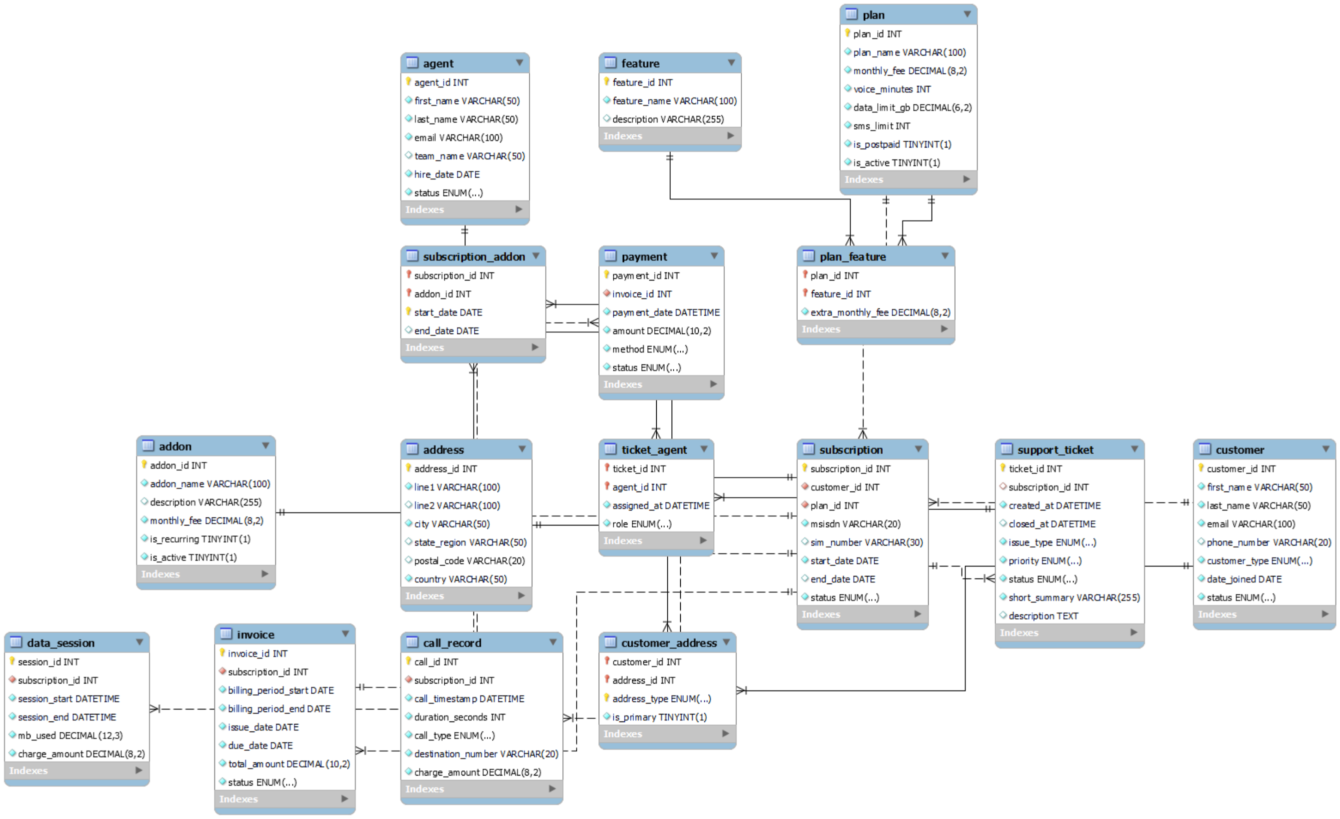
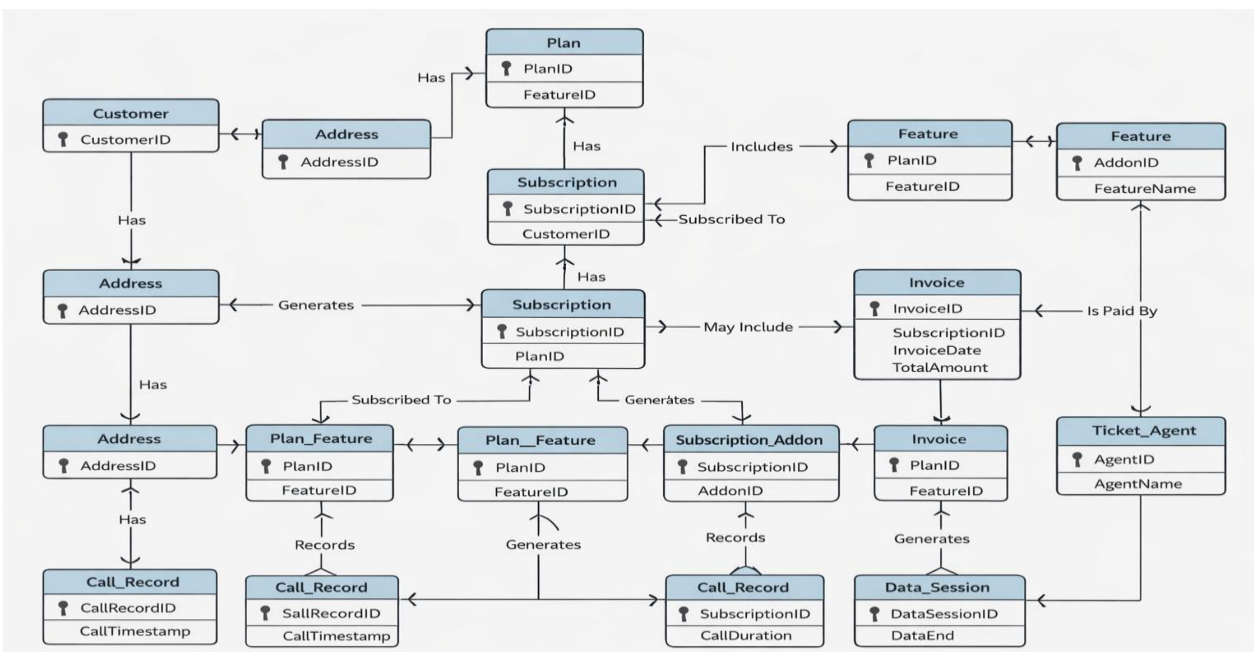


Fig: The Telecommunications Provider Database in the form of an entity-relationship diagram

The ERD is a representation of all the database entities and relationships by use of junction tables to solve many-to-many relationship and a normalized schema.



1.5 NORMALIZATION

The telecommunications database schema has been normalized to Third Normal Form (3NF) to remove redundancy, minimize update anomalies and data integrity.

Normalization rules used:

1NF: All the attributes are atomic and every record is unique.

2NF: The table is in 1NF and all the non key attributes are entirely dependent on the primary key.

3NF: The table is 2NF, and non-key attributes do not have any transitive dependencies.

Normalization Example1: Customer, Subscription, Plan:

Unnormalized design contains the customer, subscription and plan data in a single table, resulting to the repetition of customer and plan data in individual records.

1NF

The atomic values were made by eliminating repeat groups.

2NF

The entities were separated in order to eliminate partial dependencies:

Personal details are stored in customer.

Plan stores plan allowances and pricing.

A customer can be connected to a plan by subscription.

3NF

Transitive dependencies (e.g. plan prices based on subscription) were removed by making sure that such attributes are not present in any other table.

This division eliminates the duplication and update disparities.

Normalization Example 2: Plan and Feature.

Initially, several features were allowed to be stored in the same plan record, which is in 1NF because of the repetition of attributes (feature1, feature2, etc.).

1NF → 3NF

To normalize:

Unique features were put in a Feature table.

A Plan_Feature junction table was used to resolve M:N relationship.

This eliminates the occurrence of repetitive groups and allows an indefinite number of features to be added without changing the structure of tables.

Partial and transitive dependencies were verified on all the sixteen tables. All the tables represent one concept; non-key attributes only rely on primary key and all of the M:N relationships are addressed through junction tables. Thus, the end database design satisfies fully the Third Normal Form (3NF) requirements.

CHAPTER TWO: DATA POPULATION

The next thing after designing and implementing the database schema is to populate the tables with realistic sample data. The data that has been put in represents the common activities of a telecommunications company such as customer, addresses, plans, features, records of use, billing, support tickets.

SHOW TABLES;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Tables_in_db_telecom_provider
addon
address
agent
call_record
customer
customer_address
data_session
feature
invoice
payment
plan
plan_feature
subscription
subscription_addon
support_ticket
ticket_agent

Result 6

Listed all the tables created in the telecommunications schema.

1. Select * from Customer;

customer_id	first_name	last_name	email	phone_number	customer_type	date_joined	status
1	Anna	Schmidt	anna.schmidt@example.com	+491701000001	PERSONAL	2023-08-10	ACTIVE
2	Lukas	Weber	lukas.weber@example.com	+491701000002	PERSONAL	2023-09-05	ACTIVE
3	Meera	Patel	meera.patel@business.com	+491701000003	BUSINESS	2022-12-20	ACTIVE
4	Omar	Khan	omar.khan@example.com	+491701000004	PERSONAL	2024-01-11	ACTIVE
5	Sofia	Müller	sofia.mueller@example.com	+491701000005	PERSONAL	2023-03-03	ACTIVE
6	Jonas	Schneider	jonas.schneider@business.com	+491701000006	BUSINESS	2021-11-15	ACTIVE
7	Fatima	Ali	fatima.ali@example.com	+491701000007	PERSONAL	2024-02-28	ACTIVE
8	Mark	Fischer	mark.fischer@example.com	+491701000008	PERSONAL	2023-06-30	INACTIVE
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Records added in Customer table once data was populated.

2. Select * from Plan;

```
select * from Plan;
```

plan_id	plan_name	monthly_fee	voice_minutes	data_limit_gb	sms_limit	is_postpaid	is_active
1	Basic 5GB	14.99	200	5.00	100	1	1
2	Standard 20GB	29.99	500	20.00	500	1	1
3	Unlimited 100GB	49.99	9999	100.00	9999	1	1
4	Business Pro	79.99	9999	200.00	9999	1	1
5	PayG 1GB	4.99	50	1.00	50	0	1
6	Family 50GB	59.99	2000	50.00	2000	1	1
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Records in the Plan table which contains the record of available subscription plans.

3. Select subscription_id, customer_id, plan_id, msisdn, status from Subscription;

```sql
SELECT subscription_id, customer_id, plan_id, msisdn, status
FROM Subscription;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

subscription_id	customer_id	plan_id	msisdn	status
21	1	2	+491711000101	ACTIVE
22	1	5	+491711000102	ACTIVE
23	2	1	+491711000103	ACTIVE
24	3	4	+491711000104	ACTIVE
25	4	3	+491711000105	ACTIVE
26	5	6	+491711000106	ACTIVE
27	6	4	+491711000107	ACTIVE
28	7	2	+491711000108	ACTIVE
29	8	1	+491711000109	CANCELLED
30	3	6	+491711000110	ACTIVE
NULL	NULL	NULL	NULL	NULL

Record (ID 21 -30) that is used to record foreign relationships of transactional tables.

Creation of SQL Views:

Two SQL views were developed including a summary view and an analytics trend view.

View 1: Subscription Revenue Summary

Purpose:

The given perspective is a summary of the projected revenue per subscription monthly, with emphasis on high-value clients and additional revenue sources.

Usefulness for Analytics:

1. Assistance with revenue prediction.
2. Promotes value segmentation to customers.
3. Handy in the analysis of the adoption of optional add-ons.

SELECT * FROM Subscription_Revenue_Summary;

Output:

```
413 •    SELECT * FROM Subscription_Revenue_Summary;
414
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

subscription_id	first_name	last_name	plan_name	plan_fee	addon_fees	expected_monthly_revenue
23	Lukas	Weber	Basic 5GB	14.99	9.99	24.98
29	Mark	Fischer	Basic 5GB	14.99	0.00	14.99
21	Anna	Schmidt	Standard 20GB	29.99	13.98	43.97
28	Fatima	Ali	Standard 20GB	29.99	7.99	37.98
25	Omar	Khan	Unlimited 100GB	49.99	7.99	57.98
24	Meera	Patel	Business Pro	79.99	14.99	94.98
27	Jonas	Schneider	Business Pro	79.99	14.99	94.98
22	Anna	Schmidt	PayG 1GB	4.99	2.99	7.98
26	Sofia	Müller	Family 50GB	59.99	3.99	63.98
30	Meera	Patel	Family 50GB	59.99	9.99	69.98

Subscription_Revenue_Summar... ×

View 2: Monthly_Invoice_Trend

Purpose:

This trend view shows the overall revenue made out of invoices which allows the organisation to track the pattern of revenues over a period of time.

Usefulness for Analytics:

1. Monitors financial performance on a monthly basis.
2. Determines seasonal changes or fluctuations.
3. Assists in long term revenue planning and forecasting.

SELECT * FROM Monthly_Invoice_Trend;

Output:

```sql
SELECT * FROM Monthly_Invoice_Trend;
```

month	total_revenue
2024-04	494.88

CHAPTER THREE: ADVANCED SQL QUERIES IN ANALYTICS

Query 1: Top subscriptions by expected monthly revenue

The results of Query 1 indicate the best subscriptions by the amount of revenue projected after each month which was obtained by adding the monthly plan charge and the active add-on charges. Based on the findings, it is evident that business clients and those that subscribe to upper plans (like Business Pro and Family 50GB) bring the best revenue.

Output with the highest subscriptions in terms of anticipated monthly revenue and revenue segmentation:

subscription_id	customer_id	customer_name	plan_name	monthly_fee	total_addon_fee	expected_monthly_revenue	revenue_segment
24	3	Meera Patel	Business Pro	79.99	14.99	94.98	HIGH
27	6	Jonas Schneider	Business Pro	79.99	14.99	94.98	HIGH
30	3	Meera Patel	Family 50GB	59.99	9.99	69.98	HIGH
26	5	Sofia Müller	Family 50GB	59.99	3.99	63.98	HIGH
25	4	Omar Khan	Unlimited 100GB	49.99	7.99	57.98	HIGH
21	1	Anna Schmidt	Standard 20GB	29.99	13.98	43.97	MEDIUM
28	7	Fatima Ali	Standard 20GB	29.99	7.99	37.98	MEDIUM
23	2	Lukas Weber	Basic 5GB	14.99	9.99	24.98	LOW
29	8	Mark Fischer	Basic 5GB	14.99	0.00	14.99	LOW
22	1	Anna Schmidt	PayG 1GB	4.99	2.99	7.98	LOW

Query 2: Monthly invoice revenue with 3-month moving average

The result of Query 2 shows the aggregate invoice revenue during the available month (April 2024 billing based on the date of issue of invoices) and a 3-month moving average. As the sample data has only one month of invoice information, the total revenue and the moving average are the same (€494.88). Even the query manages to illustrate the application of a window function (AVG OVER) and indicate how the moving average is computed using past monthly revenue. This query would display the trend of revenues, seasonal variations, and in a real world set up that has a few months of data the query would bring a smoother understanding than the uncooked month to month values.

Output presenting the monthly total revenue and moving average of 3 months with window function:

```
SELECT month_start, total_revenue,
ROUND( AVG(total_revenue) OVER (ORDER BY mo
FROM monthly
ORDER BY month_start;
```

Result Grid | Filter Rows: | Export: | Wrap

month_start	total_revenue	moving_avg_3m
2024-04-01	494.88	494.88

Query 3: The behaviour of customers who pay and the delinquent customers

The output singles out customers who have bad or irregular payment behaviour. There are two DELINQUENT customers because of the most insignificant or no payment rates, and there are two others customers who are considered at risk with regards to incomplete payments. The fact that the latest date of payment is also in place is useful to emphasize the urgent follow-up or interventions that need to be undertaken by the billing team.

Output reveals delinquent and at-risk customers according to invoice and payment history.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	customer_name	total_invoices	total_paid	payment_rate	payment_status	last_payment_date
7	Fatima Ali	1	0.00	0.00	DELINQUENT	NULL
6	Jonas Schneider	1	40.00	0.22	DELINQUENT	2024-04-25 09:00:00
1	Anna Schmidt	2	89.94	0.53	AT_RISK	2024-04-19 16:22:00
3	Meera Patel	2	199.98	0.67	AT_RISK	2024-04-20 08:30:00

Query 4: Top agents by resolved tickets

Logic / Explanation

Number of inner queries resolved/closed tickets per agent by joining Ticket_Agent SupportTicket.

Insight: This determines the best performing support agents based on resolved tickets which could be used in operational reporting, reward schemes and staffing.

Result Grid | Filter Rows: | Export:

agent_id	agent_name	resolved_count	agent_rank
1	Sara Becker	1	1
2	Jonas Fischer	1	1

Output shows agent ranking by resolved support tickets.

Query 5: Stored Function and Procedure

Firstly, a stored function fn_customer_lifetime_value() which computes the Customer Lifetime Value (CLV) in accordance with all of the payments made by the customer in this function.

Secondly, sp_monthly_revenue_report('2024-03') which presents an accounting report on a monthly basis comprising of invoices, payments and balance.

Calculating lifetime value for a customer and Monthly Financial Summary

Zero output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

report_month	invoice_count	total_invoiced	total_paid	outstanding_amount
2024-03	0	0.00	0.00	0.00

Non-Zero output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

report_month	invoice_count	total_invoiced	total_paid	outstanding_amount
2024-04	10	1039.73	689.82	349.91

The stored operation will get the total payments of each customer so that the customer lifetime value could be compared quickly and be used to determine which customers were the high-value and low-value. The stored procedure summarizes the monthly revenue through summing invoice value and payment value in a specific month.

CHAPTER 4: APPLICATION OF CAP THEOREM IN DISTRIBUTED SYSTEMS

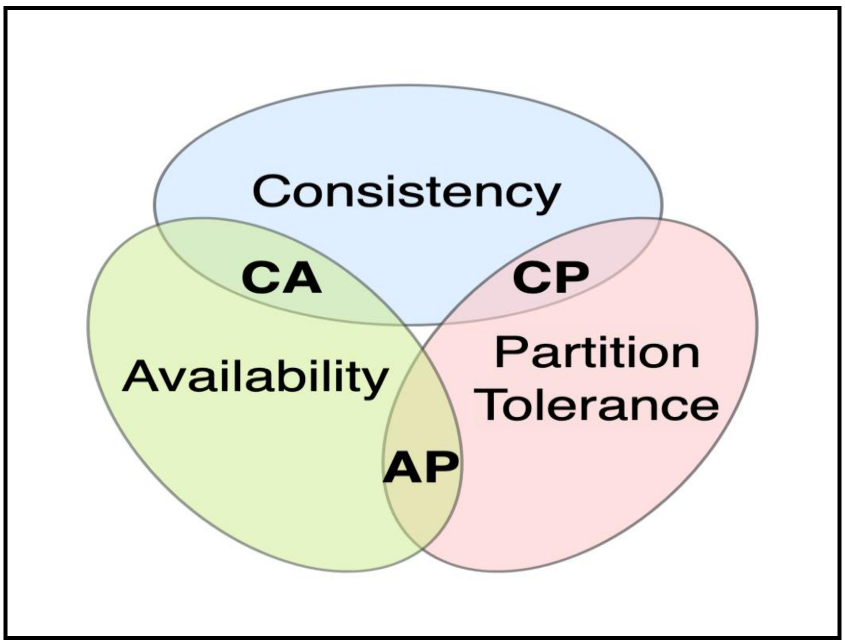
Components of CAP Theorem:

The CAP Theorem holds only two out of the three properties can be assured to a distributed database system:

Consistency (C): Each read will provide the most up-to-date and accurate data on all of the nodes in the system.

Availability (A): All requests have a response regardless of the unavailability of some nodes in the system.

Partition Tolerance (P): The system remains functional even when there are network failures that divide the communication between nodes.



The reason why all three cannot be assured at the same time:

Network partitions are inevitable in the real-world distributed systems. In case of a partition, the system should decide whether to be consistent or availed. In case of consistency, there should be some rejection of requests which will decrease availability. When the priority is on availability, the system can give out of date information, which decreases consistency. Hence, a distributed database does not provide consistency, availability and partition tolerance simultaneously.

CAP Thesis in Telecommunications:

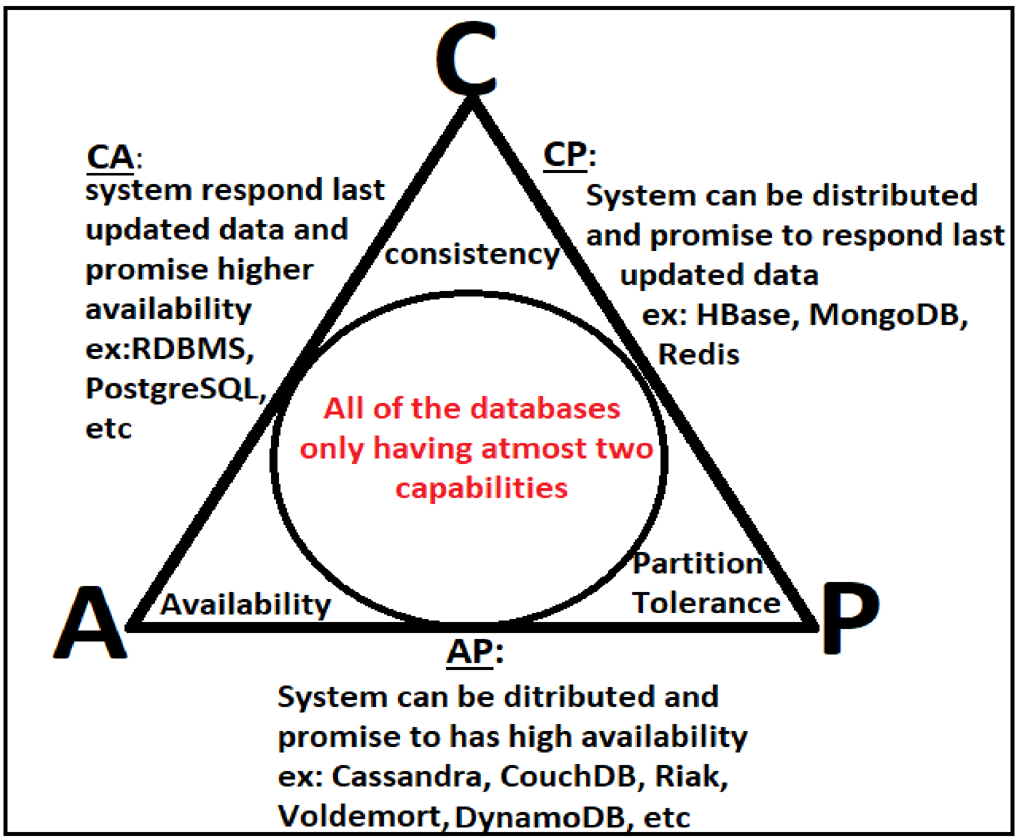
In the case of a telecommunications provider the system lays emphasis on:

Availability (A): Telecom services including voice calls, data use and provision of access to the customer should always be available.

Partition Tolerance (P): Telecom systems are geographically separated and it is therefore unavoidable for network partitions.

That which is sacrificed is Consistency (C).

Telecom systems frequently employ eventual consistency in which billing records, usage data or dashboards can be updated with a minor delay. This trade-off is satisfactory since continuous service is of greater value than instantaneous data synchronization.



CAP Effect on System Design and Scalability:

CAP theorem is a powerful factor that drives distributed database design. Data replication and eventual consistency are employed in systems that focus on availability as well as partition tolerance so that they can be scalable and tolerant to failure. The databases used by telecoms are created to be horizontal scaled, with failure tolerance and auto-recovery, even when some updates to the data might be inconsistent temporarily.

CHAPTER 5: QUERY OPTIMIZATION AND PERFORMANCE

Detection of query which is Inefficient:

The query that was identified to be potentially inefficient during analytical tasks is the customer payment behavior query (Task 3, Query 3) which has several large tables (Customer, Subscription, Invoice, Payment) and it also contains aggregations, a CASE expression and correlated subquery to find the latest payment date. This query can be slow as the database size increases and may require the repeated scans of tables and join operations on the transactional tables.

Strategies of optimization used:

Strategy 1: Indexing foreign key columns:

```
# Strategy 1- Indexing Foreign Key Columns
CREATE INDEX idx_subscription_customer ON Subscription(customer_id);
CREATE INDEX idx_invoice_subscription ON Invoice(subscription_id);
CREATE INDEX idx_payment_invoice ON Payment(invoice_id);
```

Foreign keys were included in the indexes of frequent JOINs like the customer, subscription, and the invoice. This minimizes the entire table scans as the database engine can now find matching records more effectively. Consequently, join operations are made more scalable and fast, particularly with the increase in transactional table size.

Strategy 2: Simplifying correlated query:

```
# Strategy 2
SELECT c.customer_id, CONCAT(c.first_name, ' ', c.last_name) AS customer_name, MAX(p.payment_date) AS last_payment_date
FROM Customer c
LEFT JOIN Subscription s ON c.customer_id = s.customer_id
LEFT JOIN Invoice i ON s.subscription_id = i.subscription_id
LEFT JOIN Payment p ON i.invoice_id = p.invoice_id
GROUP BY c.customer_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	customer_name	last_payment_date
1	Anna Schmidt	2024-04-19 16:22:00
2	Lukas Weber	2024-04-09 14:05:00
3	Meera Patel	2024-04-20 08:30:00
4	Omar Khan	2024-04-12 10:00:00
5	Sofia Müller	2024-04-11 11:20:00
6	Jonas Schneider	2024-04-25 09:00:00

Result 25

The correlated subquery that was used to select the last date of payment was substituted with an aggregated LEFT JOIN of the form of MAX (payment date). This optimization will decrease the number of times the sub query is calculated by calculating the result once per customer group, resulting in better performance and greater scalability as data size grows.

Performance Improvement and Reflection:

Optimization of queries minimizes table scans and I/O costs by use of indexing and simplified logic which enhances efficiency and scalability. Such optimization will guarantee low latency, cost control, reliable analytics and improved overall performance of the system in large telecom systems.

SUMMARY AND CONCLUSIONS

This assignment illustrates the design, implementation and analysis of a relational database of a telecommunications company. MySQL was used to support customer, subscription, billing, usage and support operations by having a normalized schema with realistic data, constraints, and relationship. Business insights were generated using advanced SQL queries and views, stored routines, and CAP Theorem analysis and query optimization techniques pointed out the consideration of scalability, availability, and performance in large-scale telecom systems.

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